
Learn2Design 2026: A physics experiment design competition for gravitational wave detectors

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Abstract

Learn2Design is a competition on the automated design of highly sensitive gravitational-wave detectors under realistic experimental constraints. Participants submit optimization algorithms, not fixed designs: given a detector topology, each algorithm tunes roughly 200 continuous parameters within a fixed compute budget to minimize a simulator-defined loss combining detector sensitivity and feasibility penalties. We provide the auto-differentiable simulator Differometer and approximately 30,000 high-quality designs generated through a 360,000 GPU-hour exploration campaign, which will allow for highly diverse optimization and exploration algorithms. Algorithms will be ranked by average performance on hidden topologies evaluated by the organizers, with EUR 25,000 in prize money. Beyond gravitational-wave detection, the competition asks whether AI systems can discover scientific instruments that go beyond human intuition while remaining physically meaningful.

Keywords AI4Science, Inverse Design, generative models, physics experiments, gravitational wave detector

1 Competition description

1.1 Background and impact

Physical experiments are one of the central ways in which science asks questions of nature. Modern instruments are increasingly governed by large mixed discrete-continuous design spaces: optical elements, graph topologies, component parameters, control settings, physical constraints, and performance objectives must be chosen jointly (Fig. 1). The central question addressed by this competition is therefore: **Can machine learning algorithms discover experimental designs that go beyond human intuition, while remaining physically meaningful and experimentally constrained?** We study this question in the concrete setting of interferometric gravitational-wave detectors. Gravitational waves, predicted by Einstein and directly observed for the first time in 2015, provide a new observational window into some of the most extreme events in the universe, such as black-hole mergers, neutron-star mergers, supernovae, and potentially primordial gravitational-wave sources [1–3]. The scientific reach of this field is still fundamentally limited by detector sensitivity: more sensitive detectors can observe weaker and more distant signals, improve parameter estimation, increase detection rates, and open frequency regimes in which qualitatively new astrophysics may become visible [4–7].

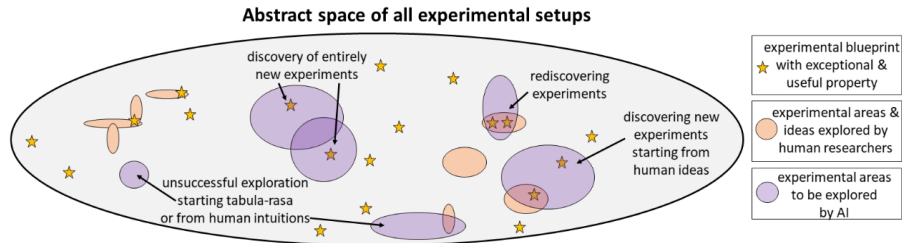


Figure 1: **Abstract search space of all experimental blueprints.** This space contains all possible experimental configurations, including those with exceptional properties (depicted by stars). Human researchers were successful, using their intuition and experience, to uncover many exceptional setups. However, many others might not have been discovered yet – and, *without help, might never be*. This space will be explored in Learn2Design.

The design of gravitational-wave detectors is a natural benchmark for AI-driven scientific discovery. An interferometric detector can be represented as a graph-like optical experiment: nodes are optical components such as mirrors, beam splitters, lasers, detectors, isolators, and quantum light sources, while edges are optical paths with continuous physical parameters (Fig. 2 and Fig. 3). The search space is both combinatorial and continuous, with strong physical constraints, non-convex objectives, expensive simulations, symmetries, feasibility boundaries, and many local optima. This makes the task scientifically realistic while matching core NeurIPS interests, including differentiable programming, generative modeling, graph learning, Bayesian optimization, reinforcement learning, evolutionary search, active learning, learned optimizers, surrogate modeling, and hybrid physics-ML methods.

Recent work has shown that AI can discover unconventional gravitational-wave detector topologies outperforming established human-designed and next-generation reference designs under realistic constraints [8]. Learn2Design turns this into a standardized competition: participants receive an open-source differentiable JAX simulator, Differometer [9], documentation, baseline algorithms, and approximately 30,000 high-quality detector designs; submitted algorithms are then run by the organizers in a standardized compute environment on hidden detector-design instances. The anticipated impact is twofold: the competition may produce new detector concepts, optimization strategies, or interpretable design motifs for gravitational-wave instrumentation, and it provides a reusable benchmark for AI-driven experiment design, where algorithms search over graphs, satisfy physical constraints, exploit differentiable simulators, and optimize a real scientific objective. Similar structures arise in quantum optics, precision sensing, microscopy, telescope design, photonic circuits, molecular design, and materials discovery [10–15].

The competition is designed for a broad NeurIPS audience, not only gravitational-wave experts. The physics is encapsulated in the simulator and scoring function, and participants can start from tutorials and more than 10 baseline optimization strategies. We expect interest from AI4Science, optimization, AutoML, differentiable programming, graph-learning, generative-modeling, reinforcement-learning, and physics-informed ML communities, with approximately 80–150 registered teams and 20–50 serious final submissions. An **unusual structural feature** is that both direct access to a differentiable utility function and a large corpus of pre-computed evaluations are available, allowing classical optimizers such as Adam, BFGS, and Bayesian optimization to compete directly with amortized methods such as prior-fitted networks.

1.2 Novelty

Learn2Design is an entirely new competition and is not a continuation of any previously accepted NeurIPS competition or recurring benchmark series. To the best of our knowledge, no previous NeurIPS competition has evaluated algorithms for the design of gravitational-wave detectors, nor more generally for the design of full scientific experiment and instrument blueprints. The competition data, hidden evaluation tasks, simulator interface, baselines, and final evaluation protocol are prepared specifically for this competition.

Several previous NeurIPS competitions are related, but none addresses the same problem. The Open Catalyst Challenge focuses on atomistic catalyst systems rather than generating new experimen-

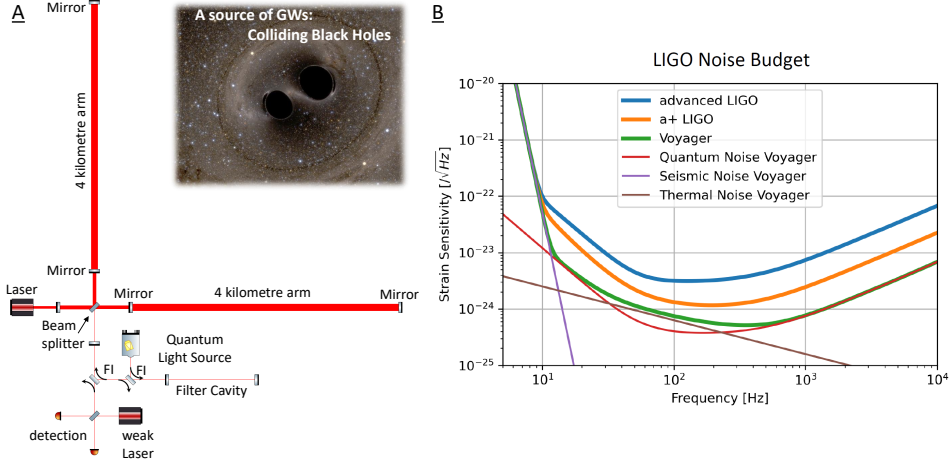


Figure 2: **Interferometric gravitational-wave detection and detector sensitivity.** Left: schematic of a laser interferometer with kilometer-scale arms, quantum light injection, and a filter cavity, together with an example astrophysical source. Right: representative strain-sensitivity curves and dominant noise contributions for Advanced LIGO, A+ LIGO, and Voyager.

tal instruments [16]; the Black-Box Optimization Challenge evaluates optimizers on generic ML hyperparameter-tuning objectives rather than physically constrained experiment design [17]; and the Machine Learning for Combinatorial Optimization competition evaluates solver decisions in discrete optimization but does not produce physical scientific devices [18]. Learn2Design combines elements of all three directions in a new setting: participants submit algorithms for discovering high-sensitivity interferometric gravitational-wave detectors under realistic constraints. A central novelty is the combination of a large solution dataset with Differometer [9], a JAX-based differentiable simulator, making the challenge accessible to Bayesian optimization, evolutionary and swarm algorithms, gradient-based optimization, graph neural networks, generative models, active learning, surrogate modeling, learned optimizers, and hybrid data-driven/simulator-based methods.

1.3 Data

The competition provides two complementary resources: a differentiable physics simulator and a large corpus of high-quality gravitational-wave detector designs. The final ranking is not based on a static test set of labels; participants submit executable algorithms evaluated by the organizers on hidden detector-design tasks. The public data make the competition accessible, enable supervised and generative learning approaches, and provide strong warm starts for active and gradient-based optimization.

Differentiable simulator. We will provide access to Differometer [9], a JAX-based auto-differentiable simulator for interferometric gravitational-wave detectors, developed in our group since 2024. It builds on the LIGO community simulator Finesse [19], developed by competition co-organizer Andreas Freise and collaborators since 1998. Given an experimental blueprint, Differometer computes a strain-sensitivity curve and experimental penalties, combined into a scalar loss. Participants can access gradients of the loss with respect to continuous design parameters, enabling gradient-based optimization, learned optimizers, active learning, surrogate modeling, Bayesian optimization with gradient information, and neural generative models followed by local refinement. The simulator is publicly available with documentation, installation instructions, examples, and starter scripts, and the organizer team includes developers of both Differometer and Finesse.

Precomputed design corpus. In addition to the simulator, we will release approximately 30,000 high-quality detector designs discovered during a six-month EuroHPC compute campaign using approximately 360,000 GPU-hours. These optimized or near-optimized examples span a broad range of parametrized design instances and optimization strategies, and can be used to learn design motifs, initialize searches, train surrogate models, or perform imitation learning. Each example will

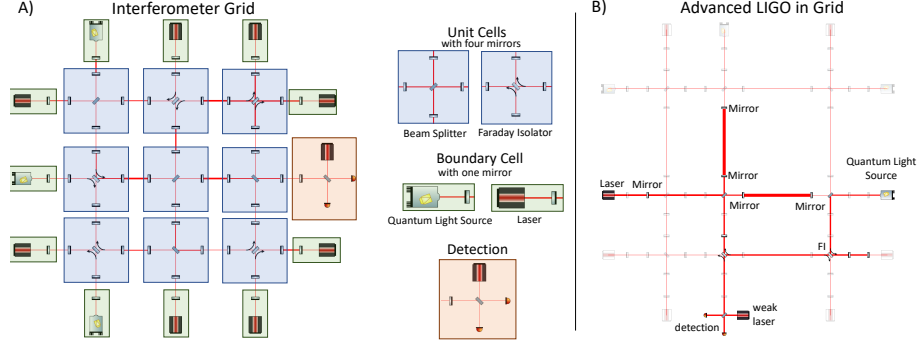


Figure 3: **Example of a large-scale interferometric Setup.** A: An interferometric grid that acts as a parametrized GW detector (also called *quasi-universal interferometer* or UIFO). Each optical element (nodes of the experimental graph) have two continuous parameters, each distance between two nodes (edges) have one parameter (their distance) B: With the right parameter setting, this Ansatz encodes the human-designed solution shown in Figure 2A. The graph-based formulation of the experimental setup allows for multiple ML approaches, including GNNs and other generative methods. (Adapted from [8])

contain, where applicable, discrete topology encodings such as topology masks, graph representations, adjacency information, and optical component types; continuous physical parameters such as mirror reflectivities, phases, optical path lengths, laser powers, squeezing parameters, or other component settings; and the scalar strain-sensitivity objective and official constraint-penalized loss. We will also provide a script to generate additional random examples with non-optimized loss values, which are cheap to produce (on the order of 0.5 seconds per example on one A100 GPU).

Availability, licensing, and leakage prevention. All public data and software will be freely available before the start of the competition. The simulator is released under the MIT licence, and baseline algorithms, submission templates, and documentation will be released through a public GitHub repository under MIT or comparable open-source licences. The documentation will describe the data format, official loss function, constraint definitions, and how additional data can be generated locally. No human data are involved; the organizers confirm compliance with all data-related concerns of the [NeurIPS Code of Ethics](#). The “ground truth” for final evaluation is not a previously published label, but is obtained only during organizer-run evaluation on hidden topologies. Public training and validation examples will be separated from hidden tasks by topology class, task family, and random seed; before release, we will remove duplicate, symmetry-equivalent, and near-duplicate designs across splits, including equivalences induced by graph isomorphisms, rotations, reflections, unused components, or other UIFO symmetries. The public leaderboard will use only public validation tasks, the final hidden leaderboard will be computed once by the organizers without iterative feedback, and participants with access to hidden tasks or final evaluation infrastructure will not be eligible to compete.

1.4 Tasks and application scenarios

Learn2Design asks participants to submit executable optimization algorithms for designing high-sensitivity interferometric gravitational-wave detectors under realistic physical constraints. Participants receive a differentiable simulator, public training data, validation tasks, documentation, and baseline algorithms. For each evaluation instance, an algorithm is given a detector-design task specifying the allowed UIFO design family, optical components, discrete topology, continuous parameter bounds, target frequency regime, and experimental constraints. Its goal is to find normalized continuous parameters in $[0, 1]$ that minimize the official loss, which combines detector sensitivity with feasibility penalties. Typical instances contain about 180–220 continuous parameters. This is an algorithmic benchmark, not a one-off design contest: participants may use simulator outputs and gradients, the 30,000-design corpus, generated training data, neural networks, surrogate models, Bayesian optimization, evolutionary search, gradient-based methods, learned optimizers, graph neural networks, generative models, or hybrid approaches. In the final evaluation, each submitted algorithm

will be run by the organizers for 4 hours on one NVIDIA A100 GPU per hidden task, on 10 different hidden topologies.

During development, participants can test on public validation tasks and compare against released baselines. In the final phase, organizers run submissions on hidden detector-design tasks separated from the public data by topology class, task family, and random seed. The official evaluator computes all candidate losses with the frozen simulator and constraint checker. The score for each task is the best verified feasible loss found within the time limit, and the final ranking is the average performance across ten hidden topologies. Additional hidden tasks may be used for top submissions when needed for statistically meaningful comparisons.

The real-world analogue is the early-stage design of future gravitational-wave detectors, where experts choose optical topologies and continuous settings such as path lengths, mirror properties, laser powers, readout choices, and squeezing parameters while satisfying hardware and feasibility constraints. A successful algorithm would act as an automated scientific design assistant, proposing promising detector concepts for expert analysis, higher-fidelity simulation, robustness checks, and possible experimental refinement. The task is challenging because the UIFO design space is mixed discrete-continuous, highly non-convex, constrained, symmetric, and full of invalid regions and local optima; the discrete topology space alone is on the order of 2^{29} . It is nevertheless tractable because participants receive Differomator, gradients, public validation tasks, scripts for generating examples, a 30,000-design archive from a six-month EuroHPC campaign, and more than 10 baseline strategies.

Although grounded in gravitational-wave instrumentation, the abstract problem is broader: designing a graph-like physical system with continuous parameters, hard constraints, and a differentiable simulator. Similar structures occur in quantum optics, integrated photonics, precision sensing, microscopy, telescope design, accelerator physics, and nanophotonic device design. The task has a positive scientific motivation and involves no human subjects, personal data, medical decisions, surveillance, weapons systems, or safety-critical deployment. The main risks are computational cost and unequal hardware access; these are mitigated through released data, runnable baselines, organizer-provided final-evaluation compute, and a capped evaluation budget.

1.5 Metrics

The official metric is the average loss achieved by a submitted optimization algorithm under a fixed compute budget. Participants submit executable algorithms, not final detector designs. In the final evaluation, the organizers run each submitted algorithm on local hardware for four hours per hidden topology, using one NVIDIA A100 GPU per run. Each hidden topology defines a fixed detector layout in the UIFO representation illustrated in Fig. 3; the underlying discrete topology space contains $4^9 \cdot 2^{11} = 2^{29}$ possible choices, and each chosen topology comes with roughly 200 normalized continuous parameters in $[0, 1]$. For each hidden topology t , the score L_t is the lowest loss found during the four-hour run, as evaluated by the official frozen Differomator simulator. The loss combines detector sensitivity with penalties for unphysical or disallowed settings, such as laser powers exceeding allowed limits. Lower loss is better. The final score is the mean loss over ten hidden topologies from a held-out hidden topology set, $S = \frac{1}{10} \sum_{t=1}^{10} L_t$. Algorithms are ranked by increasing S . The metric directly measures efficient optimization of physically constrained gravitational-wave detector designs on unseen topologies. Because all algorithms are run on the same hidden topologies, uncertainty and performance differences will be assessed using paired statistics over topologies, with paired bootstrap confidence intervals. If the leading algorithms are not statistically distinguishable after ten hidden topologies, the organizers will run them on additional hidden topologies from the same hidden set and recompute the ranking. All final evaluations will use the same frozen Differomator version, loss function, constraint checker, GPU type, time limit, and logging protocol. No subjective human judging will be used for the official ranking.

1.6 Baselines, code, and material provided

We will release a complete starting kit through the competition website and public GitHub repository at the start of the competition. It will include installation instructions for Differomator, documentation of the task format and official loss, data-loading tools for the public design corpus, scripts for generating additional random examples, example notebooks, local test scripts, and the official submission template for executable algorithms. The starting kit will include baseline optimizers covering random

search, Bayesian optimization, evolutionary algorithms, gradient-based methods such as Adam and BFGS, and learning-based warm starts such as sampling from variational autoencoders trained on the public high-quality designs. Preliminary results of several algorithms are already available on GitHub, and the remaining baselines will be prepared before the start of the competition. Each baseline will be run for 10 trials of four hours each on one NVIDIA A100 GPU 40GB VRAM, with an AMD EPYC 7302 CPU and 50 GB RAM. We will report mean loss and uncertainty across runs and development topologies. The simulator and public dataset are already available including a few selected baselines; the remaining baselines, submission templates, documentation, and benchmark tables will be released at competition launch.

1.7 Website, tutorial and documentation

The competition website will be hosted as a public GitHub repository at [Learn2Design-2026](#), serving as the central, self-contained source of information. It will contain the overview, rules, timeline, participation and submission instructions, data links, installation instructions, baseline code, evaluator code, submission templates, public leaderboards, and documentation for Differomator and the released design corpus. The front page will link to an FAQ/tutorial section with starter notebooks and scripts for installing the simulator, downloading or generating data, loading the 30,000-design corpus, evaluating candidate designs, running baselines, and packaging executable submissions. Public technical questions and discussions will be handled through GitHub Issues and GitHub Discussions; private or administrative questions will be handled through a dedicated email address. We will also link to Differomator documentation and background material describing the simulator and detector-design problem.

2 Organizational aspects

2.1 Protocol

Participation will be organized through the public GitHub repository [Learn2Design-2026](#), which has been beta-tested with local ML experts and will be tested with a larger and more diverse cohort between submission and acceptance. The repository will contain documentation, installation instructions, example submissions, baseline algorithms, scripts for running Differomator, data-download instructions, public GitHub issues/discussions, and the private email contact. Participants will not need to register to access the simulator, data, baselines, or documentation. Submissions will be executable optimization algorithms, not final detector designs. Teams register on a self-hosted submission platform and upload their solution following a template; each submission is automatically checked for installability, correct input/output format, and ability to run on a small public test instance. Participants receive automated feedback on technical failures, but not access to hidden final-evaluation tasks.

Starting in the beginning of July, participants will use the simulator, data, baselines, and public validation tasks to develop and submit new algorithms. Once per month (three times during the competition), the submitted algorithms will be evaluated by the organizers, and the results will be shared as a public leaderboard. This provides progress feedback, but no extra statistical tie-breaking. After the submission deadline, the organizers run submissions on ten hidden topologies from a held-out hidden set, with four hours on one NVIDIA A100 GPU per topology. The final ranking is the mean official Differomator loss over hidden topologies, with additional hidden topologies used for top submissions if needed for statistically meaningful rankings.

Cheating and overfitting will be limited by keeping final topologies hidden, separating public validation tasks from the final hidden set, restricting leaderboard feedback to aggregate scores, and running all final submissions ourselves in a frozen environment. Participants submit code executed by the organizers, not scores or detector parameters. Final runs use the same simulator version, loss function, constraint checker, hardware budget, timeout rules, and logging protocol for all teams, with internet disabled. To ensure fairness, final evaluation compute will be organizer-provided. Before launch, we will run dry tests with baseline submissions and external beta testers to verify the submission platform, container format, evaluator, timeout behavior, logging, and leaderboard pipeline.

2.2 Rules and Engagement

Draft contest rules. The competition is open to all participants except organizers and anyone with access to hidden evaluation tasks or final evaluation infrastructure. Participants may compete individually or in teams. Public materials can be accessed without registration; registration is only required to submit algorithms. Participants submit executable optimization algorithms, not final detector designs or manually selected hidden-task solutions. For final evaluation, each algorithm is run for four hours on one NVIDIA A100 GPU per hidden topology, across ten hidden topologies. Participants may use the public simulator, released data, baselines, generated data, and legally usable external public resources, disclosed where appropriate. They may not attempt to access hidden tasks, exploit the submission system, or use privileged information. During official evaluation, submissions must run in the organizers’ frozen environment without internet access unless identical access is enabled for all teams. Incomplete or failed submissions will simply not be ranked.

Rationale and communication. These rules ensure that the competition evaluates general optimization algorithms rather than manual tuning or overfitting. Since final submissions are run by the organizers on hidden topologies with the same simulator, loss function, hardware, and time limit, the ranking measures reproducible algorithmic performance under fair conditions, and organizer-provided compute reduces the advantage of teams with large private hardware resources. Participants will contact the organizers through GitHub issues for technical questions, a public discussion forum for general questions, and a dedicated email address for private matters. Clarifications of broad relevance will be answered publicly whenever possible. Rule updates, deadline changes, evaluator changes, and important announcements will be posted on the competition website, GitHub repository, submission platform, and participant mailing list, with all substantive changes timestamped.

2.3 Schedule and readiness

The simulator, public dataset and several baselines with their performance numbers are already available. At launch, we will release documentation, baselines, baseline performance numbers, submission templates, and instructions for submitting executable algorithms. The planned timeline is:

- **July 1, 2026:** competition starts; simulator, public data, documentation, baselines, and submission templates are released.
- **First week of August 2026:** first public leaderboard evaluation.
- **First week of September 2026:** second public leaderboard evaluation.
- **First week of October 2026:** third public leaderboard evaluation.
- **October 16, 2026:** competition closes and final submissions are due.

The public leaderboard will be released three times and used for further advertisement. The competition closes approximately eight weeks before the workshop, leaving time to run final hidden evaluations, analyze results, announce winners, and prepare the workshop program. If participation is low or technical delays occur, we will reduce or skip optional public leaderboard rounds, accept manual container submissions if the platform is unavailable, and conclude the competition using all valid final submissions under the same organizer-run hidden evaluation protocol.

2.4 Competition promotion and incentives

The competition includes a total prize pool of EUR 25,000 (approximately USD 29,000), sponsored by *SPRIND – Germany’s Federal Agency for Disruptive Innovation (Bundesagentur für Sprunginnovationen)*. The main performance prizes are:

- **First place:** EUR 10,000.
- **Second place:** EUR 6,000.
- **Third place:** EUR 3,000.

We will also award two EUR 3,000 special prizes, judged by a committee including members of the organizing team and split between two or three teams if appropriate: most surprising or creative solution, and simplest strong-performing solution. All teams whose final submissions outperform

organizer-provided baselines will be invited to contribute a short technical report, participate in a joint publication as named authors, and awarded teams will be invited to present their results during the NeurIPS Competition Track, subject to the workshop schedule.

We will promote the competition across machine learning, AI4Science, optimization, and gravitational-wave communities, including AI4Science Slack channels and mailing lists; ML4Science, optimization, AutoML, and differentiable-programming mailing lists; social media channels of the organizing team and participating institutions; targeted outreach to gravitational-wave instrumentation and adjacent scientific-design communities; SPRIND’s communication channels and deep-tech network; and under-represented communities, including through the collaboration between the Tübingen ML community and the African Institute for Mathematical Sciences (AIMS).

2.5 Competition track workshop and dissemination

We will use the NeurIPS Competition Track workshop to present the final leaderboard, baseline comparisons, uncertainty estimates, and the main algorithmic and scientific lessons from the competition. The best-performing teams, including winners of the main and special prizes, will be invited to present through talks, lightning talks, or posters, depending on the workshop format. We will also organize discussion time between participants, organizers, and gravitational-wave domain experts to address reproducibility, interpretation of discovered designs, and lessons for AI-driven scientific experiment design.

After the competition, we will publish the final results, evaluation protocol, baselines, and analysis on the competition GitHub repository. We do not plan a separate call for papers; instead, teams with high-performing algorithms or exceptionally surprising, creative, or scientifically informative solutions will be invited to collaborate on a joint competition-review paper as co-authors.

3 Resources

3.1 Organizing team

The organizing team combines machine learning, AI-driven experimental design, and gravitational-wave detector physics. Philipp Hennig and his group provide expertise in optimization, probabilistic modeling, Bayesian optimization, active learning, and ML competitions. Mario Krenn and the Artificial Scientist Lab coordinate the scientific vision, AI-for-experimental-design aspects, simulator interface, public dataset, and AI4Science outreach. Jonathan Klimesch, Laurin Sefa, Soham Basu, Priya Kanagasabapathi, Sören Arlt, Xuemei Gu, Thomas Christie, and Colin Doumont will support the simulator, data release, baselines, submission templates, platform testing, and technical evaluation pipeline. Andreas Freise, and Rana Adhikari provide LIGO, Virgo, KAGRA, and Einstein Telescope expertise to ensure physically meaningful tasks, constraints, and interpretation.

Jonathan Klimesch will coordinate the competition with strong support from Mario Krenn, Philipp Hennig, and the rest of the team. Laurin Sefa and the Tübingen ML teams will provide baselines and starter code. The submission platform and evaluator will be tested before launch, with additional beta tests from ML and gravitational-wave collaborators. Final evaluations will be run by the organizers in the standardized compute environment. The team is diverse in discipline, career stage, and institution, spanning machine learning, automated scientific discovery, quantum and optical simulation, and gravitational-wave instrumentation.

3.2 Resources provided by organizers

The organizer team at the University of Tuebingen will run the submitted algorithms on A100 GPUs using its dedicated machine learning cluster and administrative support. SPRIND will provide the EUR 25,000 prize money.

3.3 Support requested

The winning teams need to present their work live at the conference. It would help us to know the number of available complimentary tickets. We would also like to take part in any travel fellowship scheme for early-career researchers or colleagues from underrepresented communities or countries.

A Biography of all team members

- **Jonathan Klimesch**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Jonathan Klimesch is a PhD student in the Artificial Scientist Lab at the University of Tübingen. He developed the JAX-based Differometer simulator, and was the main organizer of the 360k GPU-hours EuroHPC run for obtaining 30,000 high-quality solutions – these are the two data sources in the competition.
- **Laurin Sefa**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Laurin Sefa is an undergraduate student of Machine Learning at the University of Tuebingen. He implemented more than 10 baselines to provide diverse, immediate starting points for participants in the competition.
- **Soham Basu**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Soham Basu is a PhD student in the Artificial Scientist Lab at the University of Tübingen, with a strong background in Bayesian optimization and meta-optimization. He developed code to improve the usability of the competition software.
- **Priya Kanagasabapathi**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Priya Kanagasabapathi is a PhD student in the Artificial Scientist Lab at the University of Tübingen, with a background in fine-tuning LLMs for optimization and AutoML. She develops agentic systems for automated discovery of physics experiments and physics concepts.
- **Sören Arlt**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Sören Arlt is a PhD student in the Artificial Scientist Lab at the University of Tübingen, with a background in automated discovery of new quantum experiments and language models for conceptual discoveries in physics.
- **Xuemei Gu**, Institut für Festkörpertheorie und Optik, Friedrich-Schiller-Universität Jena, Jena, Germany.
Xuemei Gu leads the AI Quantum Lab at the University of Jena. She is an expert in developing differentiable simulators for quantum physics and applying them to the automated design of quantum physics experiments.
- **Thomas Christie**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Thomas Christie is a PhD student in the Methods of Machine Learning group at the University of Tübingen. His research focuses on developing scalable uncertainty-aware models and leveraging them for principled decision-making in tasks such as Bayesian optimisation and active learning. He is particularly interested in applying these methods to challenging scientific applications, such as optimising the properties of materials and molecules.
- **Colin Doumont**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Colin Doumont is a PhD student in the Methods of Machine Learning group at the University of Tübingen. His research focuses on probabilistic models (e.g. Gaussian processes or diffusion models) and sequential decision-making (e.g. Bayesian optimisation), aiming to find simple solutions that lead to strong performance. He is interested in applying these methods to real-world problems, particularly the natural sciences.
- **Andreas Freise**, Nikhef, National Institute for Subatomic Physics, Amsterdam, The Netherlands.
Andreas Freise is Professor of Gravitational Wave Physics at Vrije Universiteit Amsterdam and Nikhef, and Project Director for the international Einstein Telescope organization. He is a leading expert in laser interferometry and gravitational-wave instrumentation, has played major roles in LIGO, Virgo, and the Einstein Telescope, and initiated the development of FINESSE, one of the most widely used interferometer simulation tools in the gravitational-wave community.

- **Rana Adhikari**, Institute for Quantum Information and Matter, California Institute of Technology, Pasadena, CA, USA.
Rana Adhikari is a professor of physics at the California Institute of Technology (Caltech) and leads Caltech’s LIGO Laboratory. Rana works on the experimental physics of gravitational wave detection and is among the scientists responsible for the US-based Laser Interferometer Gravitational wave Observatory (LIGO) that first observed gravitational waves in 2015. Rana has received the *New Horizons in Physics Prize 2019* for his contributions in the design of next-generation gravitational wave detectors.
- **Philipp Hennig**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Philipp Hennig is a full professor of Machine Learning at the University of Tübingen, Germany, and Director of the Tübingen AI Center. His team develops methods and algorithms across the domains of machine learning, from optimization to simulation and probabilistic functionality. Philipp has co-organized multiple NeurIPS workshops and established the International Conference of Probabilistic Numerics. With partners at Google, Meta, and others, he co-initiated the AlgoPerf Competition at MLCommons, whose results advanced the training of large deep architectures and affected recent frontier LLMs.
- **Mario Krenn**, Department for Computer Science, Faculty of Science, University of Tuebingen, Tuebingen, Germany.
Mario Krenn is a full professor for Machine Learning in Science at the University of Tübingen, Germany, and leads the Artificial Scientist Lab. His team has developed automated design methodologies for physics problems since 2016 [20], with applications ranging from quantum optics to gravitational-wave instrumentation. Together with Rana Adhikari and Yehonathan Drori, he developed AI methods for discovering novel interferometric gravitational-wave detector topologies [8]. He created and organized the 2021 IEEE Big Data Cup Challenge “Science4Cast”, and will coordinate the competition’s scientific vision, AI-for-discovery framing, and outreach to the AI4Science community.

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